

# Monte Carlo Methods

Introduction

Generating Samples

**Markov Chain Monte Carlo + Discrepancy**

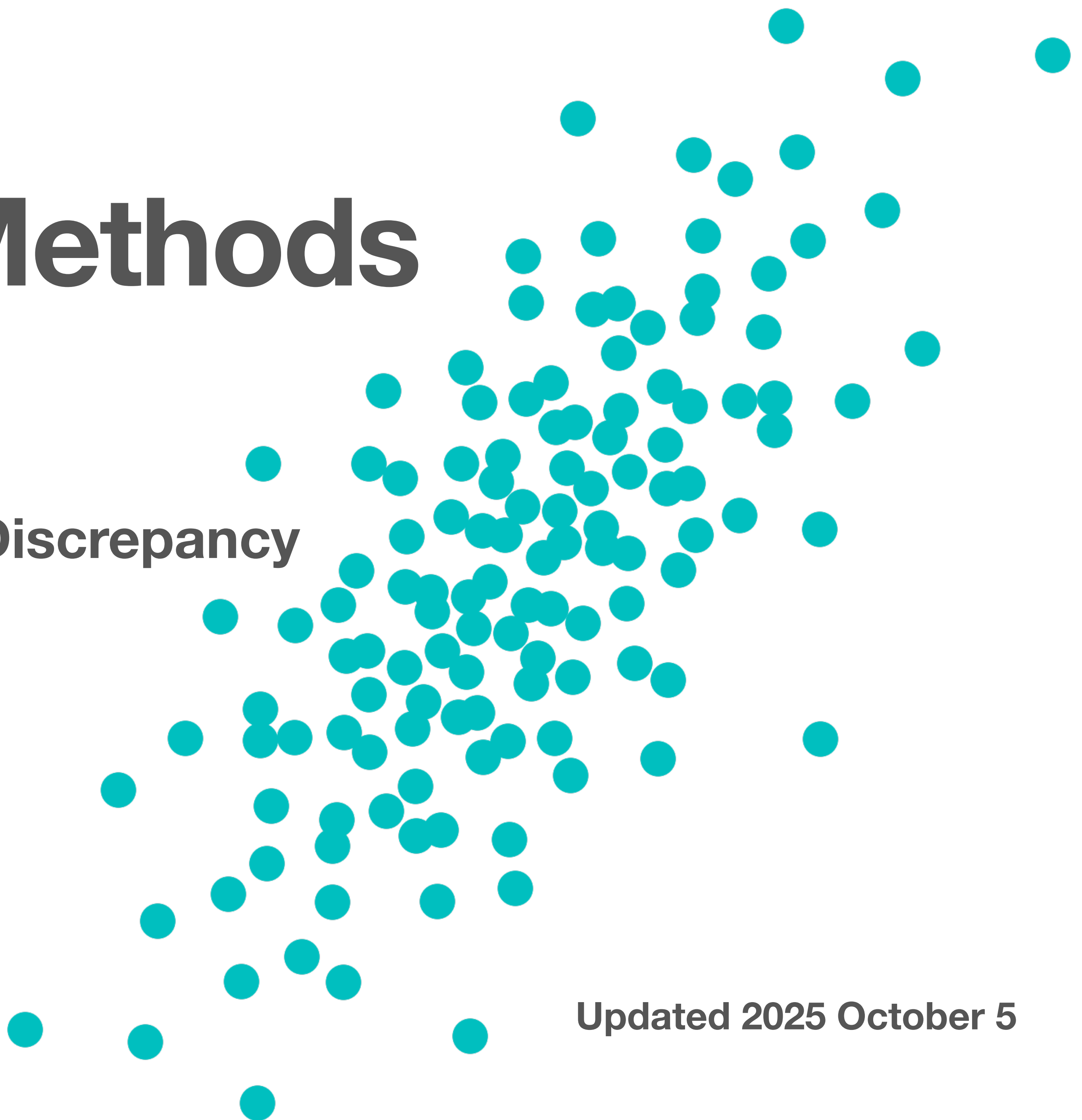
Improving Efficiency

Selected Topics

Git [website](#) and [repository](#)  
Canvas

Fred Hickernell, Fall 2025

Updated 2025 October 5

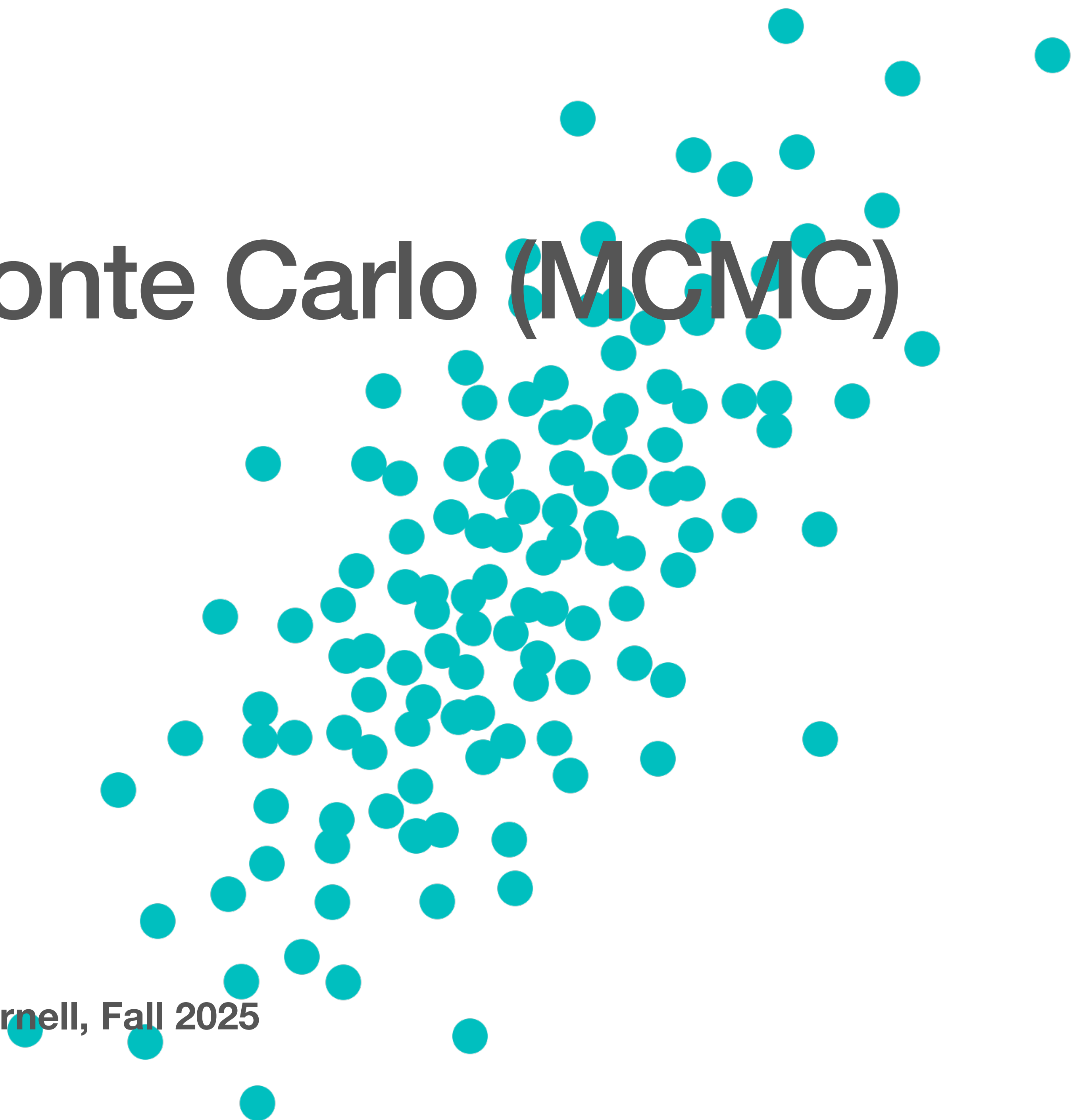


# Markov Chain Monte Carlo (MCMC) + Discrepancy

Owen, Chapters 11– 12

Project Selection due Oct 1

Assignment 3 due Oct 17





# Rate your confidence heading into Test 1

- Go to [menti.com](https://www.menti.com)
- Use code 4923 2283





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  - An unnormalized target density



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  - An unnormalized target density
  - May reject a lot of samples



# How do we generate samples when the distribution is complicated?

- **Inverse cumulative distribution** function applied to uniform samples works for simple cases
- **Correlated multivariate Gaussian** distributions are possible through affine transformations of univariate Gaussians
- **Acceptance-rejection** sampling
  - Propose from a simple density that you know bounds
  - An unnormalized target density
  - May reject a lot of samples
- **Markov chain Monte Carlo** (MCMC) moves samples toward areas of higher target density





# Markov Chains

Let  $\mathcal{X}$  be a state space for a **sequence** of random variables. Markov chains follow a probability law satisfying

$$\mathbb{P}(X_{n+1} \in \mathcal{A} \mid X_n = x_n, \dots, X_0 = x_0) = \mathbb{P}(X_{n+1} \in \mathcal{A} \mid X_n = x_n)$$

where  $\mathcal{A}$  is some (measurable) subset of  $\mathcal{X}$ . This means that **where you go next only depends on where you are now**, not how you got there.

We use these to

- Generate  $X_0, X_1, \dots$  whose **probability distribution** converges to the target
- Study **queuing** systems

We are often interested in the **asymptotic** behavior of the  $X_i$



# Metropolis–Hastings algorithm

Suppose that

- You really **want to** sample  $X$  with known (unnormalized) **target** density  $q$
- You have a **proposal** density  $q_{\text{new}|\text{old}}$  to select the next point starting from where you are

Given  $X[0]$

For  $i = 0$  to  $n-1$

Generate  $Z \sim q_{\text{new}|\text{old}}(\cdot | X_i)$ ,  $U \sim \mathcal{U}[0,1]$

If  $U \leq \min\left(1, \frac{q(Z)q_{\text{new}|\text{old}}(X_i | Z)}{q(X_i)q_{\text{new}|\text{old}}(Z | X_i)}\right) = \min\left(1, \frac{\text{joint density of first } Z \text{ then } X_i}{\text{joint density of first } X_i \text{ then } Z}\right)$

$X[i+1] \leftarrow Z$

Else

$X[i+1] \leftarrow X[i]$

Note that you **always accept** either the new or the old



# Metropolis algorithm

If  $q_{\text{new}|\text{old}}(x | z) = q_{\text{new}|\text{old}}(z | x)$ , then this can be simplified

Metropolis sampling proceeds as follows

Given  $X[0]$

For  $i = 0$  to  $n-1$

Generate  $Z \sim q_{\text{new}|\text{old}}(\cdot | X_i)$ ,  $U \sim \mathcal{U}[0,1]$

If  $U \leq \min\left(1, \frac{q(Z)}{q(X_i)}\right)$

$X[i+1] \leftarrow Z$

Else

$X[i+1] \leftarrow X[i]$





# Advantages of and Challenges in MCMC

- Samples from complicated, unnormalized densities
- Easy to implement
- Proposal density and starting point need tuning to ensure that
  - Explore all the sample space and not miss out
  - Exploit sampling in regions where the density is large





# How do we measure quality of MCMC

Short answer: We cannot easily

Longer answer: Given a symmetric, positive definite kernel,  $K : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ , the **discrepancy**, sometimes called the maximum mean discrepancy between two empirical distributions is defined by

$$D^2(\{\mathbf{x}_i\}_{i=0}^{m-1}, \{\mathbf{z}_i\}_{i=0}^{n-1}; K) = \frac{1}{m^2} \sum_{i,j=0}^{m-1} K(\mathbf{x}_i, \mathbf{x}_j) - \frac{2}{mn} \sum_{i,j=0}^{m-1, n-1} K(\mathbf{x}_i, \mathbf{z}_j) + \frac{1}{n^2} \sum_{i,j=0}^{n-1} K(\mathbf{z}_i, \mathbf{z}_j)$$

Requires  $\mathcal{O}(\max(m, n)^2)$  operations





# What is discrepancy?

The **discrepancy** measures

- The **difference** between two probability distributions
- The **worst-case error** in approximating an expectation/integral  $\mu = \mathbb{E}[f(X)]$  by a sample mean when  $f$  is in the unit ball of a Hilbert space with **reproducing kernel**  $K$
- The **root-mean square error** in approximating an expectation/integral  $\mu = \mathbb{E}[f(X)]$  by a sample mean when  $f$  is an instance of a Gaussian process with **covariance kernel**  $K$

Discrepancies from kernels were popularized by [Hickernell 1998] and [Gretton et al. 2012]



# Symmetric, Positive Definite Kernels

- A square matrix,  $K$ , is symmetric and positive definite iff

$$K^T = K \quad \text{and} \quad \mathbf{c}^T K \mathbf{c} > 0 \text{ for all } \mathbf{c} \neq \mathbf{0}$$

- A kernel,  $K : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ , is **symmetric and positive definite** iff

$$K := \left( K(\mathbf{x}_i, \mathbf{x}_j) \right)_{i,j=0}^{n-1}, \quad \text{the Gram matrix, is symmetric and positive definite}$$

for all  $n \in \mathbb{N}$  and  $\{\mathbf{x}_i\}_{i=0}^{n-1}$  with distinct elements, e.g.

$$K(\mathbf{t}, \mathbf{x}) = \exp\left(-\|\mathbf{t} - \mathbf{x}\|^2/h^2\right), \quad \mathcal{X} = \mathbb{R}^d, \quad \text{squared exponential}$$

$$K(\mathbf{t}, \mathbf{x}) = \prod_{\ell=1}^d \left[ 1 + \frac{\gamma_{\ell}^2}{2} \left( |t_{\ell} - 1/2| + |x_{\ell} - 1/2| - |t_{\ell} - x_{\ell}| \right) \right], \quad \mathcal{X} = [0,1]^d$$

centered discrepancy



# Discrepancy from kernels

Suppose that

- $K : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$  is a symmetric, positive definite kernel,
- $\mathcal{X}$  is the sample space for random variables/vectors  $\mathbf{X}$  and  $\mathbf{Z}$
- $\mathbf{X}$  has CDF  $F_X$  and  $\mathbf{Z}$  has CDF  $F_Z$

Then **discrepancy** measures the difference between two distributions:

$$\begin{aligned} D^2(F_X, F_Z; K) &= \mathbb{E}_{\mathbf{X}, \mathbf{X}' \sim F_X} K(\mathbf{X}, \mathbf{X}') - 2\mathbb{E}_{\mathbf{X} \sim F_X, \mathbf{Z} \sim F_Z} K(\mathbf{X}, \mathbf{Z}) + \mathbb{E}_{\mathbf{Z}, \mathbf{Z}' \sim F_Z} K(\mathbf{Z}, \mathbf{Z}') \\ &= \int_{\mathcal{X} \times \mathcal{X}} K(\mathbf{x}, \mathbf{x}') dF_X(\mathbf{x}) dF_X(\mathbf{x}') - 2 \int_{\mathcal{X} \times \mathcal{X}} K(\mathbf{x}, \mathbf{z}) dF_X(\mathbf{x}) dF_Z(\mathbf{z}) \\ &\quad + \int_{\mathcal{X} \times \mathcal{X}} K(\mathbf{z}, \mathbf{z}') dF_Z(\mathbf{z}) dF_Z(\mathbf{z}') \geq 0 \end{aligned}$$





# Discrepancy from kernels

Then **discrepancy** measures the difference between two distributions:

$$\begin{aligned} D^2(F_X, F_Z; K) &= \int_{\mathcal{X} \times \mathcal{X}} K(\mathbf{x}, \mathbf{x}') dF_X(\mathbf{x}) dF_X(\mathbf{x}') - 2 \int_{\mathcal{X} \times \mathcal{X}} K(\mathbf{x}, \mathbf{z}) dF_X(\mathbf{x}) dF_Z(\mathbf{z}) \\ &\quad + \int_{\mathcal{X} \times \mathcal{X}} K(\mathbf{z}, \mathbf{z}') dF_Z(\mathbf{z}) dF_Z(\mathbf{z}') \\ &= \int_{\mathcal{X} \times \mathcal{X}} K(\mathbf{x}, \mathbf{x}') d[F_X(\mathbf{x}) - F_Z(\mathbf{x})] d[F_X(\mathbf{x}') - F_Z(\mathbf{x}')] \\ &= \int_{\mathcal{X} \times \mathcal{X}} K(\mathbf{x}, \mathbf{x}') \varrho_X(\mathbf{x}) \varrho_X(\mathbf{x}') d\mathbf{x} d\mathbf{x}' - 2 \int_{\mathcal{X} \times \mathcal{X}} K(\mathbf{x}, \mathbf{z}) \varrho_X(\mathbf{x}) \varrho_Z(\mathbf{z}) d\mathbf{x} d\mathbf{z} \\ &\quad + \int_{\mathcal{X} \times \mathcal{X}} K(\mathbf{z}, \mathbf{z}') \varrho_Z(\mathbf{z}) \varrho_Z(\mathbf{z}') d\mathbf{z} d\mathbf{z}' \geq 0 \end{aligned}$$



# Discrepancy with empirical distributions

The formula for the squared discrepancy works if one or both of the distributions is the **empirical distribution** of a sample.

$$\begin{aligned} D^2(F_X, F_{\{z_i\}_{i=0}^{n-1}}; K) &= D^2(F_X, \{z_i\}_{i=0}^{n-1}; K) \\ &= \int_{\mathcal{X} \times \mathcal{X}} K(x, x') dF_X(x) dF_X(x') - \frac{2}{n} \sum_{i=0}^{n-1} \int_{\mathcal{X}} K(x, z_i) dF_X(x) \\ &\quad + \frac{1}{n^2} \sum_{i,j=0}^{n-1} K(z_i, z_j) \end{aligned}$$

$$D^2(\{x_i\}_{i=0}^{m-1}, \{z_i\}_{i=0}^{n-1}; K) = \frac{1}{m^2} \sum_{i,j=0}^{m-1} K(x_i, x_j) - \frac{2}{mn} \sum_{i,j=0}^{m-1, n-1} K(x_i, z_j) + \frac{1}{n^2} \sum_{i,j=0}^{n-1} K(z_i, z_j)$$



# Unbiased estimates of squared discrepancy

The squared discrepancy is non-negative, but we can construct **unbiased and possibly negative** estimates of  $D^2(F_X, F_Z; K)$  based on IID samples  $\{\mathbf{x}_i\}_{i=0}^{m-1}$  and/or  $\{\mathbf{z}_i\}_{i=0}^{n-1}$

$$D_{\text{unb}}^2(F_X, \{\mathbf{z}_i\}_{i=0}^{n-1}; K) = \int_{\mathcal{X} \times \mathcal{X}} K(\mathbf{x}, \mathbf{x}') d\varrho_X(\mathbf{x}) d\varrho_X(\mathbf{x}') - \frac{2}{n} \sum_{i=0}^{n-1} \int_{\mathcal{X}} K(\mathbf{x}, \mathbf{z}_i) d\varrho_X(\mathbf{x}) \\ + \frac{1}{n(n-1)} \sum_{\substack{i,j=0 \\ i \neq j}}^{n-1} K(\mathbf{z}_i, \mathbf{z}_j)$$

$$D_{\text{unb}}^2(\{\mathbf{x}_i\}_{i=0}^{m-1}, \{\mathbf{z}_i\}_{i=0}^{n-1}; K) = \frac{1}{m(m-1)} \sum_{\substack{i,j=0 \\ i \neq j}}^{m-1} K(\mathbf{x}_i, \mathbf{x}_j) - \frac{2}{mn} \sum_{i,j=0}^{m-1, n-1} K(\mathbf{x}_i, \mathbf{z}_j) \\ + \frac{1}{n(n-1)} \sum_{\substack{i,j=0 \\ i \neq j}}^{n-1} K(\mathbf{z}_i, \mathbf{z}_j)$$





# Discrepancy as worst case error

Let  $K : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$  be a **symmetric, positive definite** kernel. Then

- $\mathcal{H} :=$  **completion** (add limits of Cauchy sequences) of
$$\{c_0 K(\cdot, \mathbf{x}_0) + \cdots + c_{n-1} K(\cdot, \mathbf{x}_{n-1}) : n \in \mathbb{N}, c_i \in \mathbb{R}, \mathbf{x}_i \in \mathcal{X}\}$$
is a **Hilbert space** (vector space + inner product) of functions defined on  $\mathcal{X}$  with **reproducing kernel**  $K$
- Which means  $f(\mathbf{x}) = \langle K(\cdot, \mathbf{x}), f \rangle_{\mathcal{H}}$  for all  $f \in \mathcal{H}, \mathbf{x} \in \mathcal{X}$
- Suppose that  $\int_{\mathcal{X} \times \mathcal{X}} K(\mathbf{x}, \mathbf{x}') dF_X(\mathbf{x}) dF_X(\mathbf{x}')$  is defined and finite, which means that  $f \mapsto \int_{\mathcal{X}} f(\mathbf{x}) dF_X(\mathbf{x})$  is a **bounded linear functional** on  $\mathcal{H}$
- This means that **cubature error**,  $f \mapsto \int_{\mathcal{X}} f(\mathbf{x}) dF_X(\mathbf{x}) - n^{-1} \sum_{i=0}^{n-1} f(\mathbf{z}_i)$  is also a bounded linear functional on  $\mathcal{H}$





# Discrepancy as worst case error (cont'd)

- **Reproducing property:**  $f(\mathbf{x}) = \langle K(\cdot, \mathbf{x}), f \rangle_{\mathcal{H}}$  for all  $f \in \mathcal{H}$ ,  $\mathbf{x} \in \mathcal{X}$
- **Cubature error,**  $f \mapsto \int_{\mathcal{X}} f(\mathbf{x}) dF_{\mathbf{X}}(\mathbf{x}) - n^{-1} \sum_{i=0}^{n-1} f(\mathbf{z}_i)$ , is a bounded linear functional on  $\mathcal{H}$
- By the **Riesz Representation Theorem** there exists a  $\zeta \in \mathcal{H}$  such that

$$\int_{\mathcal{X}} f(\mathbf{x}) dF_{\mathbf{X}}(\mathbf{x}) - \frac{1}{n} \sum_{i=0}^{n-1} f(\mathbf{z}_i) = \langle \zeta, f \rangle_{\mathcal{H}} \quad \forall f \in \mathcal{H}$$

- The **Cauchy-Schwarz inequality** separates the contribution to the error from  $\{\mathbf{z}_i\}_{i=0}^{n-1}$  and from  $f$

$$\left| \int_{\mathcal{X}} f(\mathbf{x}) dF_{\mathbf{X}}(\mathbf{x}) - \frac{1}{n} \sum_{i=0}^{n-1} f(\mathbf{z}_i) \right| = \left| \langle \zeta, f \rangle_{\mathcal{H}} \right| \leq \|\zeta\|_{\mathcal{H}} \|f\|_{\mathcal{H}} \quad \forall f \in \mathcal{H}$$



# Discrepancy as worst case error (cont'd)

$$\left| \int_{\mathcal{X}} f(\mathbf{x}) \, dF_X(\mathbf{x}) - \frac{1}{n} \sum_{i=0}^{n-1} f(\mathbf{z}_i) \right| = \left| \langle \zeta, f \rangle_{\mathcal{H}} \right| \leq \|\zeta\|_{\mathcal{H}} \|f\|_{\mathcal{H}} \quad \forall f \in \mathcal{H}$$

- By the reproducing property and the Riesz Representation Theorem,

$$\zeta(\mathbf{x}') = \langle K(\cdot, \mathbf{x}'), \zeta \rangle_{\mathcal{H}} = \int_{\mathcal{X}} K(\mathbf{x}, \mathbf{x}') \, dF_X(\mathbf{x}) - \frac{1}{n} \sum_{i=0}^{n-1} K(\mathbf{z}_i, \mathbf{x}')$$

$$\begin{aligned} \|\zeta\|_{\mathcal{H}}^2 = \langle \zeta, \zeta \rangle_{\mathcal{H}} &= \int_{\mathcal{X} \times \mathcal{X}} K(\mathbf{x}, \mathbf{x}') \, dF_X(\mathbf{x}) dF_X(\mathbf{x}') - \frac{2}{n} \sum_{i=0}^{n-1} \int_{\mathcal{X}} K(\mathbf{x}, \mathbf{z}_i) \, dF_X(\mathbf{x}) \\ &\quad + \frac{1}{n^2} \sum_{i,j=0}^{n-1} K(\mathbf{z}_i, \mathbf{z}_j) = D^2(F_X, \{\mathbf{z}_i\}_{i=0}^{n-1}; K) \end{aligned}$$



# Discrepancy as worst case error (cont'd)

In particular, for **uniform target** distributions

$$\left| \int_{[0,1]^d \times [0,1]^d} f(\mathbf{x}) \, d(\mathbf{x}) - \frac{1}{n} \sum_{i=0}^{n-1} f(\mathbf{z}_i) \right| \leq D(\mathcal{U}[0,1]^d, \{\mathbf{z}_i\}_{i=0}^{n-1}; K) \|f\|_{\mathcal{H}} \quad \forall f \in \mathcal{H}$$

where

$$\begin{aligned} D^2(\mathcal{U}[0,1]^d, \{\mathbf{z}_i\}_{i=0}^{n-1}; K) = & \int_{[0,1]^d \times [0,1]^d} K(\mathbf{x}, \mathbf{x}') \, d\mathbf{x} d\mathbf{x}' - \frac{2}{n} \sum_{i=0}^{n-1} \int_{[0,1]^d} K(\mathbf{x}, \mathbf{z}_i) \, d\mathbf{x} \\ & + \frac{1}{n^2} \sum_{i,j=0}^{n-1} K(\mathbf{z}_i, \mathbf{z}_j) \end{aligned}$$





# After Test 1, what do we do differently?

- Go to [menti.com](https://www.menti.com)
- Use code 8510 7177
- Two slides
  - One to help me know how I can help
  - One for you to reflect on what you will do differently



# My response to your feedback

- You requested more worked examples, especially those that mimic test problems
  - You need to study to learn **concepts**, not to be able to do problems
  - I will do **more examples** in class
  - I will assign more problems for homework of the **type** that may appear on tests (see the revised Assignment 3)
  - I encourage you to look at **old tests**
  - My tests **will not intentionally copy** class examples or old test problems



# My response to your feedback

- You asked for more background material and were concerned that the class required advanced mathematics or statistics.
- The statistics and mathematics required on the tests are a calculus and a calculus-level probability class
  - How to take **expectations**
  - The binomial, exponential, and normal/Gaussian **distributions**
  - **Sample** means and variances
  - **Confidence** intervals
- I do not expect you to know the details of functional analysis or stochastic processes, but to understand the overall argument
- To help me understand what advanced mathematics or statistics you need help with please respond at [mentimeter.com](https://www.mentimeter.com) 2644 2309





# My response to your feedback

- There were comments about the test format, how to recover from a low score and number of tests
  - I need to have **in-person** tests to gauge what you have learned
  - I will take the **highest** of your two test scores rather than average the two
  - This is a **graduate** class suitable for masters and doctoral students





# Discrepancy as average case error

Suppose that

- $f$  is drawn from a Gaussian process with zero mean and **covariance kernel**  $K : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ , i.e.,  $f \sim \mathcal{GP}(0, K)$ .
- Do not need to specify the sample space for  $f$ , which will almost surely have less smoothness than the Hilbert space with reproducing kernel  $K$ .

The mean squared error for a deterministic cubature rule is

$$\begin{aligned} & \mathbb{E}_{f \in \mathcal{GP}(0, K)} \left| \int_{\mathcal{X}} f(\mathbf{x}) \, dF_{\mathbf{X}}(\mathbf{x}) - \frac{1}{n} \sum_{i=0}^{n-1} f(\mathbf{z}_i) \right|^2 \\ &= \int_{\mathcal{X} \times \mathcal{X}} \mathbb{E}_{f \in \mathcal{GP}(0, K)} [f(\mathbf{x}) f(\mathbf{x}')] \, dF_{\mathbf{X}}(\mathbf{x}) dF_{\mathbf{X}}(\mathbf{x}') - \frac{2}{n} \sum_{i=0}^{n-1} \int_{\mathcal{X}} \mathbb{E}_{f \in \mathcal{GP}(0, K)} [f(\mathbf{x}) f(\mathbf{z}_i)] \, dF_{\mathbf{X}}(\mathbf{x}) \\ & \quad + \frac{1}{n^2} \sum_{i, j=0}^{n-1} \mathbb{E}_{f \in \mathcal{GP}(0, K)} [f(\mathbf{z}_i) f(\mathbf{z}_j)] \end{aligned}$$



# Discrepancy as average case error (cont'd)

The mean squared error for a deterministic cubature rule is

$$\begin{aligned} & \mathbb{E}_{f \in \mathcal{GP}(0, K)} \left| \int_{\mathcal{X}} f(\mathbf{x}) \, dF_{\mathbf{X}}(\mathbf{x}) - \frac{1}{n} \sum_{i=0}^{n-1} f(\mathbf{z}_i) \right|^2 \\ &= \int_{\mathcal{X} \times \mathcal{X}} \mathbb{E}_{f \in \mathcal{GP}(0, K)} [f(\mathbf{x}) f(\mathbf{x}')] \, dF_{\mathbf{X}}(\mathbf{x}) dF_{\mathbf{X}}(\mathbf{x}') - \frac{2}{n} \sum_{i=0}^{n-1} \int_{\mathcal{X}} \mathbb{E}_{f \in \mathcal{GP}(0, K)} [f(\mathbf{x}) f(\mathbf{z}_i)] \, dF_{\mathbf{X}}(\mathbf{x}) \\ & \quad + \frac{1}{n^2} \sum_{i, j=0}^{n-1} \mathbb{E}_{f \in \mathcal{GP}(0, K)} [f(\mathbf{z}_i) f(\mathbf{z}_j)] \\ &= \int_{\mathcal{X} \times \mathcal{X}} K(\mathbf{x}, \mathbf{x}') \, dF_{\mathbf{X}}(\mathbf{x}) dF_{\mathbf{X}}(\mathbf{x}') - \frac{2}{n} \sum_{i=0}^{n-1} \int_{\mathcal{X}} K(\mathbf{x}, \mathbf{z}_i) \, dF_{\mathbf{X}}(\mathbf{x}) + \frac{1}{n^2} \sum_{i, j=0}^{n-1} K(\mathbf{z}_i, \mathbf{z}_j) \\ & \quad = D^2(F_{\mathbf{X}}, \{\mathbf{z}_i\}_{i=0}^{n-1}) \end{aligned}$$



# Observations about discrepancy

- The value of the discrepancy depends on the **choice of kernel**,  $K$ , and the parameters that define it
  - $K$  may be chosen for convenience
  - Should choose the **hyperparameters** of  $K$  to ensure that your  $f$  is in the **middle**; this can be done by **empirical Bayes/maximum likelihood estimation** in the average case setting
  - $K$  may be chosen based on knowledge about **properties** of  $f$ , such as domain, smoothness, periodicity, importance of different coordinates
  - $D(\cdot, \cdot; c^2 K) = |c| D(\cdot, \cdot; K)$
- $D(F_X, \{z\}_{i=0}^{n-1}; K)$  **measures the quality** of  $\{z\}_{i=0}^{n-1}$  for estimating the mean,  $\mu = \mathbb{E}[f(X)]$ , by the sample mean  $\hat{\mu}_n = n^{-1} \sum_{i=0}^{n-1} f(z)$  for
  - $f$  in a Hilbert space,  $\mathcal{H}$ , with reproducing kernel  $K$  (don't need to know  $\|\cdot\|_{\mathcal{H}}$  explicitly) or
  - $f \sim \mathcal{GP}(0, K)$





# Observations about discrepancy (cont'd)

- The mean square discrepancy of an **IID random** sample is

$$\mathbb{E}\left[D^2\left(F_{\mathbf{X}}, \{z_i \stackrel{\text{IID}}{\sim} F_{\mathbf{X}}\}_{i=0}^{n-1}; K\right)\right] = \frac{1}{n} \left[ \int_{\mathcal{X}} K(\mathbf{x}, \mathbf{x}) dF_{\mathbf{X}}(\mathbf{x}) - \int_{\mathcal{X} \times \mathcal{X}} K(\mathbf{x}, \mathbf{x}') dF_{\mathbf{X}}(\mathbf{x}) dF_{\mathbf{X}}(\mathbf{x}') \right]$$

- The deterministic cubature error bound **cannot** be used constructively to bound the error because the norm of  $f$  cannot be estimated
- If  $f \sim \mathcal{GP}(0, K)$ , and the parameters of  $K$  are estimated by empirical Bayes (maximum likelihood), then one can construct **credible intervals** for the cubature error

$$\mathbb{P}_{f \in \mathcal{GP}(0, K)} \left[ \left| \int_{\mathcal{X}} f(\mathbf{x}) dF_{\mathbf{X}}(\mathbf{x}) - \frac{1}{n} \sum_{i=0}^{n-1} f(z_i) \right| \leq \frac{2.58 D(F_{\mathbf{X}}, \{z_i\}_{i=0}^{n-1}; K)}{\sqrt{n}} \right] \geq 99 \%$$



# Example of the Centered discrepancy

$$K(t, \mathbf{x}) = \prod_{\ell=1}^d \left[ 1 + \frac{\gamma_{\ell}^2}{2} (|t_{\ell} - 1/2| + |x_{\ell} - 1/2| - |t_{\ell} - x_{\ell}|) \right], \quad \mathcal{X} = [0, 1]^d$$

$$\|f\|_{\mathcal{H}}^2 = \|f(0.5, \dots, 0.5)\|_2^2$$

$$\begin{aligned} & + \left\| \frac{\partial f(x_1, 0.5, \dots, 0.5)}{\gamma_1 \partial x_1} \right\|_2^2 + \left\| \frac{\partial f(0.5, x_2, 0.5, \dots, 0.5)}{\gamma_2 \partial x_2} \right\|_2^2 + \dots \\ & + \left\| \frac{\partial f(x_1, x_2, 0.5, \dots, 0.5)}{\gamma_1 \gamma_2 \partial x_1 \partial x_2} \right\|_2^2 + \left\| \frac{\partial f(x_1, 0.5, x_3, 0.5, \dots, 0.5)}{\gamma_1 \gamma_3 \partial x_1 \partial x_3} \right\|_2^2 + \dots \\ & + \dots + \left\| \frac{\partial f(x_1, \dots, x_d)}{\gamma_1 \cdots \gamma_d \partial x_1 \cdots \partial x_d} \right\|_2^2 \end{aligned}$$



# And now back to MCMC





# Maximum likelihood estimation (MLE)

Let  $\theta$  be a parameter to be estimated, and let

$q_{\text{DATA}}(\text{data}|\theta)$  = density for random DATA given parameter  $\theta$   
= also known as **likelihood( $\theta|\text{data}$ )**

$\theta_{\text{MLE}}$  maximizes **likelihood( $\theta$  | data)** is called the **maximum likelihood estimate**

E.g. if the data is  $y_0, \dots, y_{n-1} \stackrel{\text{IID}}{\sim} \mathcal{N}(\mu, \sigma^2)$ , then

$$q_{\text{DATA}}(\{y_i\}_{i=0}^{n-1} | (\mu, \sigma^2)) = \frac{1}{(2\pi\sigma)^{n/2}} \exp\left(-\sum_{i=0}^{n-1} (y_i - \mu)^2\right)$$

$$\mu_{\text{MLE}} = \frac{1}{n} \sum_{i=0}^{n-1} y_i, \quad \sigma_{\text{MLE}}^2 = \frac{1}{n} \sum_{i=0}^{n-1} (y_i - \mu_{\text{MLE}})^2$$





# Bayesian Inference

$\varrho_{\text{prior}}(\theta)$  = density expressing **prior** belief about parameter  $\Theta$  at  $\theta$

$\varrho_{\text{DATA}|\Theta}(\text{data}|\theta)$  = density for DATA given  $\Theta = \theta$  evaluated at data  
= also know as **likelihood** $_{\Theta|\text{DATA}}(\theta|\text{data})$

Then

$\varrho_{\text{posterior}}(\theta|\text{data})$  = density expressing belief about  $\Theta$  given data and prior belief

$$\begin{aligned} &= \frac{\varrho_{\text{joint}}(\theta, \text{data})}{\varrho_{\text{DATA}}(\text{data})} = \frac{\varrho_{\text{DATA}|\Theta}(\text{data}|\theta) \varrho_{\text{prior}}(\theta)}{\varrho_{\text{DATA}}(\text{data})} \\ &= \frac{\text{likelihood}_{\Theta|\text{DATA}}(\theta|\text{data}) \varrho_{\text{prior}}(\theta)}{\varrho_{\text{DATA}}(\text{data})} \propto \text{likelihood}_{\Theta|\text{DATA}}(\theta|\text{data}) \varrho_{\text{prior}}(\theta) \end{aligned}$$

Have **unnormalized** posterior, i.e. can ignore anything depending only on data

May compute maximum a posteriori (MAP) or posterior mean of  $\theta$  through MCMC sampling



# Overcoming challenges of MCMC



# Queues

Imagine a customer service system with random arrival times of customers and random service times. There may be several stages. We consider the case where each  $i$  represents the next event. E.g.,

$A_i$  = arrival time of next customer  $\overset{\text{IID}}{\sim} F_A$

$N_i$  = number of customers in the queue

$S_i$  = service time of first customer in queue  $\overset{\text{IID}}{\sim} F_S$

Given  $A[0] \sim F_A$ ,  $N[0] \leftarrow 0$ ,  $S[0] \sim F_S$

For  $i = 0$  to  $n-1$

    If  $N[i] = 0$    #get new customer

$N[i+1] \leftarrow 1$ ,  $A[i+1] \sim F_A$ ,  $S[i+1] \leftarrow S[i]$

    ElseIf  $A[i] < S[i]$    #get new customer

$N[i+1] \leftarrow N[i+1] + 1$ ,  $A[i+1] \sim F_A$ ,  $S[i+1] \leftarrow S[i] - A[i]$

    Else   #finish serving customer

$N[i+1] \leftarrow N[i+1] - 1$ ,  $A[i+1] = A[i] - S[i]$ ,  $S[i+1] \sim F_S$





# Queues

Imagine a customer service system with random arrival times of customers and random service times. There may be several stages. Suppose that you want to measure customer waiting time and total time

$A_i$  = arrival time of next customer  $\stackrel{\text{IID}}{\sim} F_A$

$N_i$  = number of customers in the queue

$S_i$  = service time of first customer in queue  $\stackrel{\text{IID}}{\sim} F_S$

$T_i$  = total time passed

$W_j$  = waiting time plus service time for the  $j^{\text{th}}$  customer



# Queues

Given  $A[0] \sim F_A$ ,  $N[0] \leftarrow 0$ ,  $S[0] \sim F_S$ ,  $T[0] \leftarrow 0$

$F \leftarrow 0$  #index of first customer in queue

$E \leftarrow 0$  #index of next customer to enter queue

For  $i = 0$  to  $n-1$

    If  $N[i] = 0$  #get new customer

$N[i+1] \leftarrow 1$ ,  $A[i+1] \sim F_A$ ,  $S[i+1] \leftarrow S[i]$ ,  $T[i+1] \leftarrow T[i+1] + A[i]$

$W[E] \leftarrow A[i]$ ,  $E \leftarrow E+1$

    ElseIf  $A[i] < S[i]$  #get new customer

$N[i+1] \leftarrow N[i+1] + 1$ ,  $A[i+1] \sim F_A$ ,  $S[i+1] \leftarrow S[i] - A[i]$ ,  $T[i+1] \leftarrow T[i+1] + A[i]$

$W[E] \leftarrow A[i]$ ,  $W[E-1] \leftarrow W[E-1] + A[i]$ , ...,  $W[F] \leftarrow W[F] + A[i]$ ,  $E \leftarrow E+1$

    Else #finish serving customer

$N[i+1] \leftarrow N[i+1] - 1$ ,  $A[i+1] = A[i] - S[i]$ ,  $S[i+1] \sim F_S$ ,  $T[i+1] \leftarrow T[i+1] + S[i]$

$W[E-1] \leftarrow W[E-1] + S[i]$ , ...,  $W[F] \leftarrow W[F] + S[i]$ ,  $F \leftarrow F+1$